County-Level Risk Stratification for Managed Care Optimization

Kelly Kroeper 2026

Geographic Health Risk Is Uneven — Yet Intervention Models Remain Uniform

- Preventable utilization risk demonstrates geographic concentration.
- Clinical burden and social vulnerability exhibit structural misalignment.
- Uniform intervention strategies fail to capture regional risk heterogeneity.

National averages obscure localized structural risk.

Composite Risk Model Architecture

Chronic Burden Domain

- Diabetes
- Hypertension
- Obesity
- COPD

SDOH Risk Domain

- Food insecurity
- Housing instability
- Transportation access
- Social isolation

Percentile Normalization

0-100

Weighted Composite Model

- 60% Chronic
- 40% SDOH

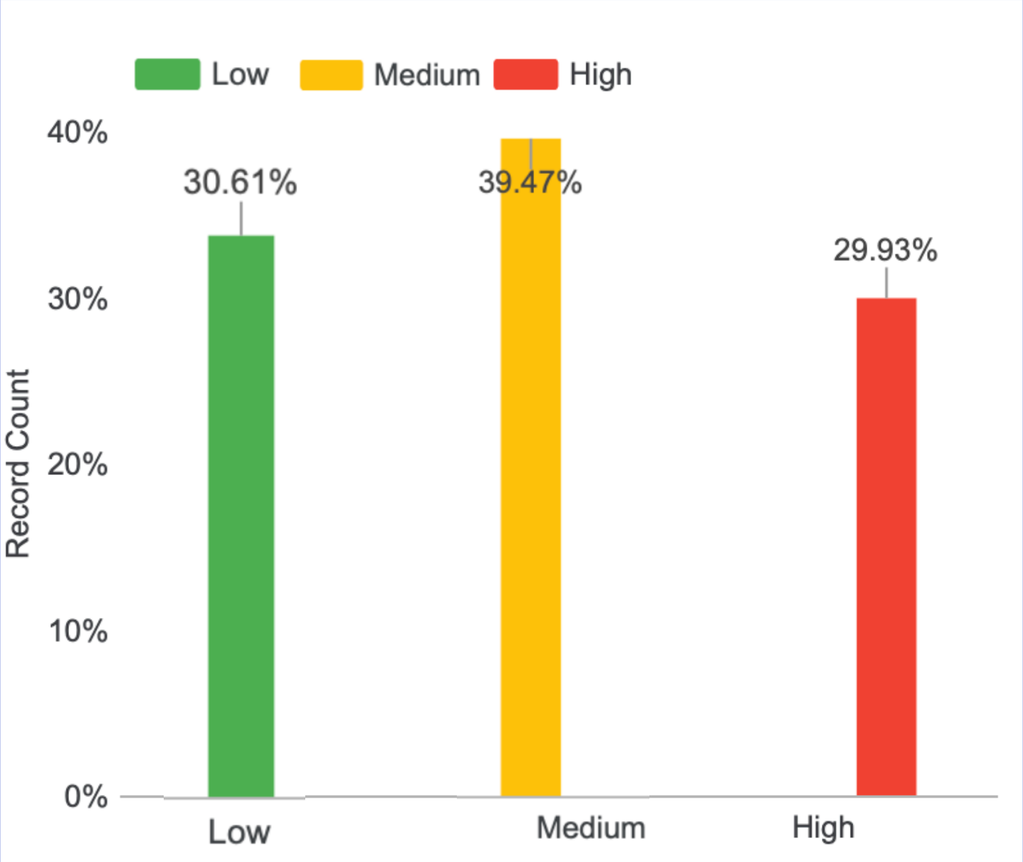
Risk-Tier Segmentation

High (Top 30%)
Medium (Middle 40%)
Low (Bottom 30%)

$$\text{Composite Risk Score} = 0.6C + 0.4S$$

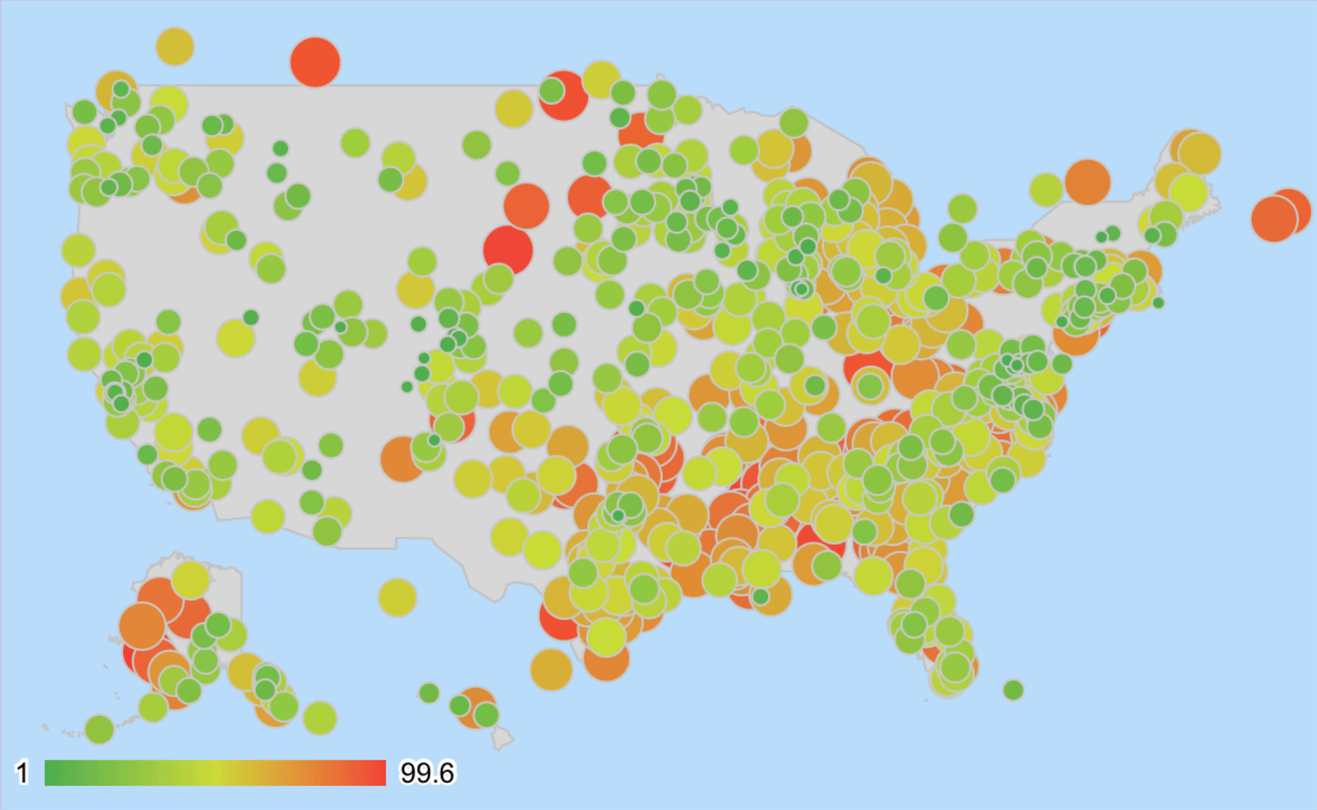
C = Chronic Percentile S = SDOH Percentile

High Composite Risk Affects Nearly One-Third of U.S. Counties



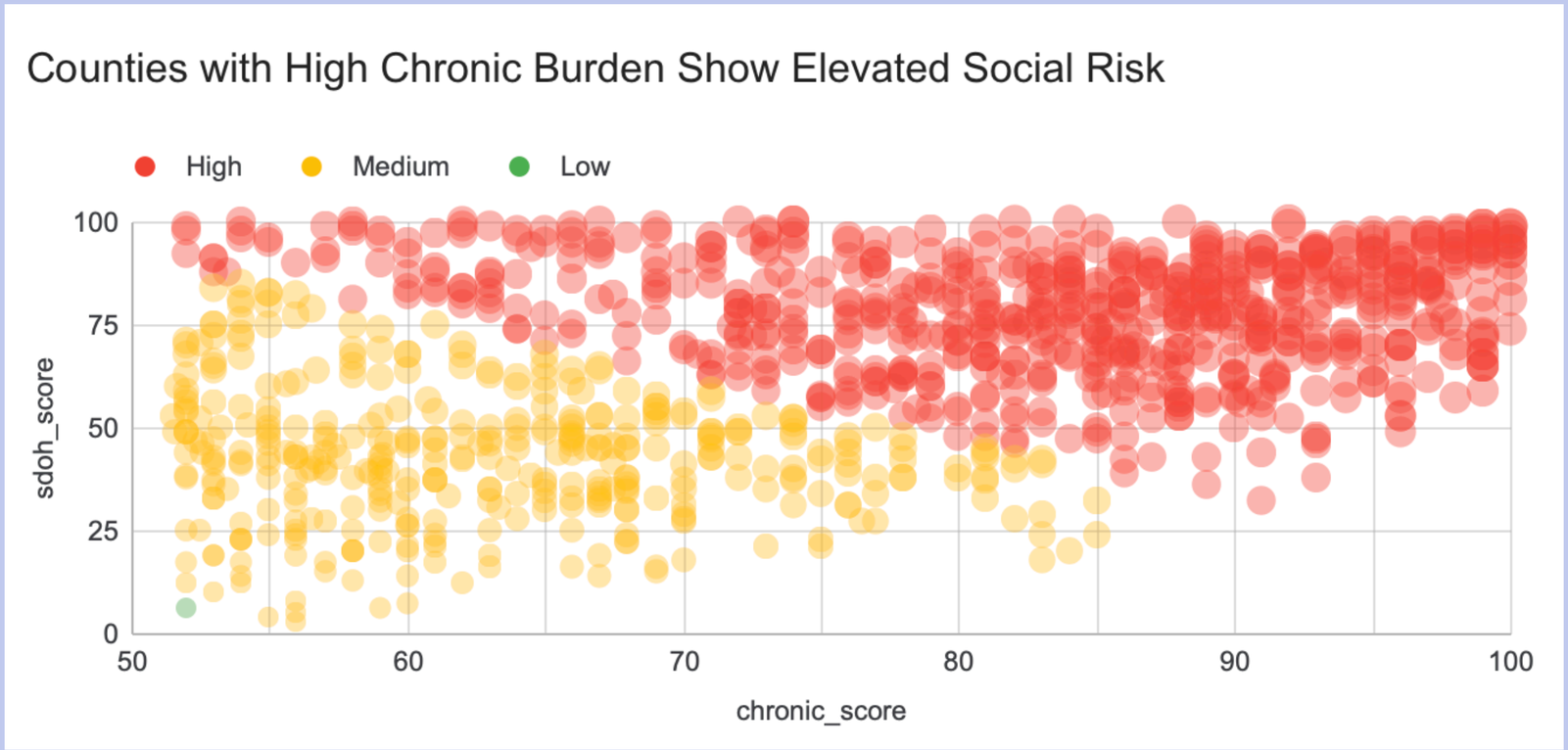
High composite risk represents a material share of the national geography, reflecting structural exposure rather than isolated outliers.

High Composite Risk Demonstrates Clear Regional Clustering



High-risk counties demonstrate visible regional clustering patterns rather than random dispersion. Geographic concentration suggests scalable intervention zones for managed care deployment.

Risk Convergence and Divergence



- Moderate positive correlation (r ≈ 0.50)
- 22.6-point mean chronic–SDOH divergence
- 28% of counties exhibit severe misalignment (>30 points)

Operational Implications for Managed Care

Geographic Care Targeting

- Prioritize high composite-risk counties for care management expansion.
- Deploy community-based interventions in structurally aligned high-burden regions.
- Inform market-level resource allocation decisions.

Structural Misalignment Detection

- Identify counties where social vulnerability outpaces clinical burden.
- Detect potential access gaps or underdiagnosed populations.
- Target preventive outreach before chronic escalation.

Value-Based Strategy Alignment

- Support value-based contract prioritization.
- Inform provider network adequacy planning.
- Enable region-specific intervention models instead of uniform deployment.

Geographic segmentation enables precision intervention — reducing reliance on uniform national strategies.

Next Phase: Scaling the Risk Model

Claims & Utilization Integration

- Integrate claims-based cost and utilization data.
- Validate composite risk score against actual utilization patterns.
- Quantify ROI of geographic targeting.

Predictive Layer Development

- Transition from descriptive risk segmentation to predictive modeling.
- Incorporate longitudinal trend analysis.
- Evaluate time-based risk escalation.

Market-Level Drilldown

- Enable MSA- or plan-level segmentation.
- Overlay Medicare Advantage / Medicaid penetration data.
- Support contract-level performance monitoring.

Intervention Tracking Framework

- Track outcomes pre- and post-intervention.
- Monitor care gap closure rates.
- Evaluate SDOH program impact by geography.

This framework establishes a scalable foundation for geographic risk intelligence within managed care environments.